

Supplementary Material 2

Article Title

Assessment of occurrence, diversity, and biomass of macroinvertebrates in Swiss groundwater systems using citizen science data

Running head

Macroinvertebrates in Swiss groundwater systems

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Detailed procedure for macroinvertebrate bio-area measurements

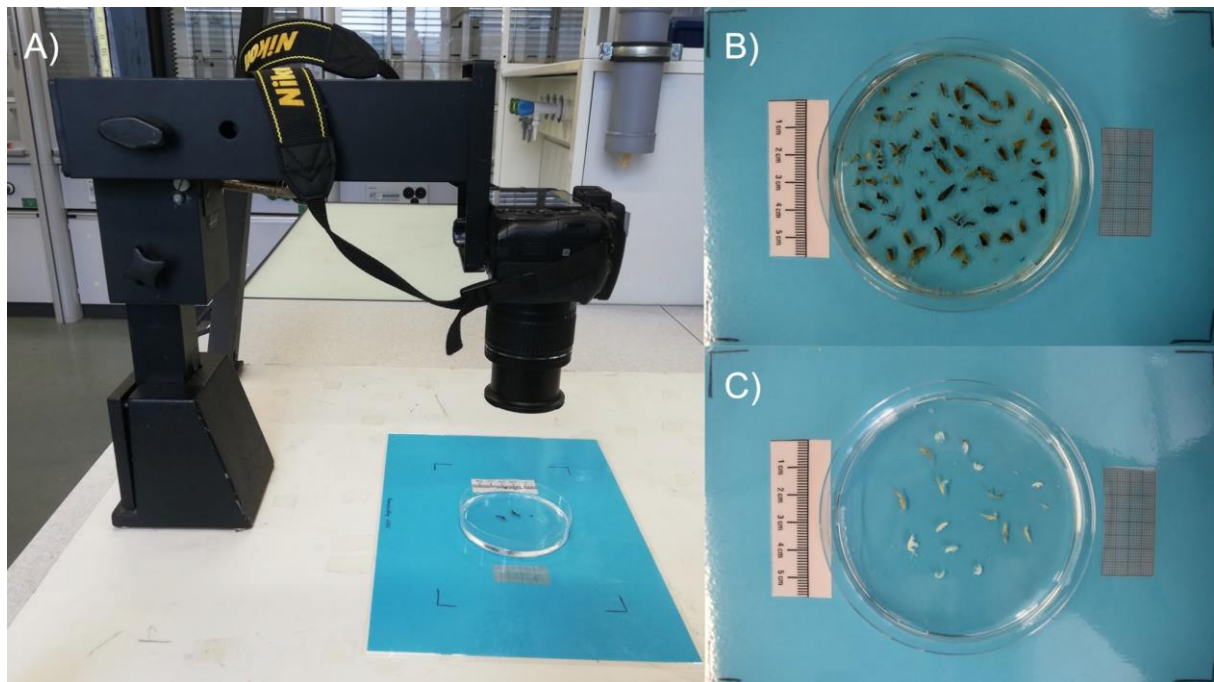


Figure S2. Set up for obtaining macroinvertebrate pictures for bio-area calculation. All macroinvertebrates from one sample were placed in a petri dish over a blue paper to contrast with both, pigmented and unpigmented specimens (A-C).

ImageJ Batch Processing code used to obtain area and intensity of all organisms in each sample

```
// Batch Processing Code to calculate area and intensity of all
macroinvertebrates of all pictures
// Author: Mara Knüsel, 15.02.22; Edited: Ana Sofia Schneider & Mara
Knüsel,
24.02.23
// input field shows directory of input pictures
// output directory field left empty (output directory programmed in code)
original_picture_name = getTitle();
output_dir = "/Thesis/2_Data/2_biomass_pictures/2_output_imageJ/"
//create copy of original picture for using in the end as control picture
run("Duplicate...", "title=controlpic2.jpg");
controlpic2=getTitle();
//create copy of original picture for using in color thresholding
run("Duplicate...", "title=thresholdpic.jpg");
thresholdpic_name=getTitle();
// set the scale by marking the scale on the image and indicating its value
selectWindow(thresholdpic_name);
//setTool("line");
makeLine(4830, 1750, 4821, 2065);
run("Set Scale...", "distance=304.8235 known=10 unit=mm global");
// Color Thresholding based on Red, Green, Blue
min=newArray(3);
max=newArray(3);
filter=newArray(3);
a=getTitle();
call("ij.plugin.frame.ColorThresholder.RGBtoYUV");
run("RGB Stack");
run("Convert Stack to Images");
selectWindow("Red");
```

```

rename("0");
selectWindow("Green");
rename("1");
selectWindow("Blue");
rename("2");
min[0]=0;
max[0]=255;
filter[0]="pass";
min[1]=0;
max[1]=255;
filter[1]="pass";
min[2]=0;
max[2]=108;
filter[2]="stop";
for (i=0;i<3;i++){
    selectWindow(""+i);
    setThreshold(min[i], max[i]);
    run("Convert to Mask");
    if (filter[i]=="stop") run("Invert");
}
imageCalculator("AND create", "0","1");
imageCalculator("AND create", "Result of 0","2");
for (i=0;i<3;i++){
44
    selectWindow(""+i);
    close();
}
selectWindow("Result of 0");
close();
selectWindow("Result of Result of 0");
rename(thresholdpic_name);
// Colour Thresholding finished-----
// Create first Duplicate for control picture showing marked animals
run("Duplicate...", "title=thresholdpic2.jpg");
thresholdpic2_name = getTitle();
// Create second Duplicate for control picture with removed background
selectWindow(thresholdpic_name);
run("Duplicate...", "title=thresholdpic3.jpg");
thresholdpic3_name = getTitle();
// Calculate Area of all animals using the thresholded picture and obtain
pigmentation values (mean grey value)
redirectImageTitle = original_picture_name;
selectWindow(thresholdpic_name);
// define relevant area (petri dish) and set size threshold = 1
//setTool("oval");
makeOval(1704, 730, 2541, 2596);
run("Set Measurements...", "area mean limit redirect=" + redirectImageTitle
+ " decimal=3");
run("Analyze Particles...", "size=1-Infinity show=[Overlay Masks] display
clear add");
run("Labels...", "color=red font=36 show draw bold");
saveAs("Results.csv", output_dir + original_picture_name+"_Measurements" +
".csv");
saveAs("jpeg", output_dir +
original_picture_name+"_control_picture_numbered" + ".jpg");
// Create control picture showing chosen animals marked in red
selectWindow(thresholdpic2_name);
run("8-bit");
run("Colors...", "foreground=white background=red selection=orange");
run("Create Selection");
selectWindow(original_picture_name);
run("Restore Selection");
run("Clear", "slice");

```

```

selectWindow(original_picture_name);
run("RGB Color");
//setTool("oval");
makeOval(1704, 730, 2541, 2596);
run("Draw", "slice");
saveAs("jpeg", output_dir + original_picture_name+"_control_picture" +
".jpg");
// Create control picture with background removed
selectWindow(thresholdpic3_name);
run("8-bit");
run("Colors...", "foreground=red background=white selection=orange");
run("Create Selection");
run("Make Inverse");
selectWindow(controlpic2);
run("Restore Selection");
run("Clear", "slice");
45
selectWindow(controlpic2);
run("RGB Color");
//setTool("oval");
makeOval(1704, 730, 2541, 2596);
run("Draw", "slice");
saveAs("jpeg", output_dir + original_picture_name
+"_control_picture_inverse" + ".jpg");
close("*");

```

The obtained images were first thresholded based on Red, Green, and Blue (RGB) components to differentiate the organisms from the background. Then the mean grey value and the area of all organisms were measured through particle analysis. Particles with an area smaller than 1 pixel² were excluded and all other particles were overlaid with a unique colored mask and were assigned a numeric label (Fig. S3).



Figure S3. Example of the output obtained from the batch processing in ImageJ. The three images on the left were created for visually controlling the thresholding process and the CSV file containing the area measurements and the mean grey values of all organisms of a sample was processed in R.

Procedure for *Niphargus*, *Gammarus*, and oligochaetes

Area and pigmentation measurements for *Niphargus*, *Gammarus*, and oligochaetes were compiled and added separately to the final dataset, as these specimens were previously sorted out for further work. For Oligochaeta we used the obtained area and pigmentation measurements from an exemplary Oligochaeta specimen and added these values to the bio-area of samples initially containing oligochaetes. To obtain an area measurement for the *Gammarus* specimens, we photographed 21 *Gammarus* specimens (10 *G. fossarum* A, 7 *G. fossarum* B, and 4 *G. pulex*) and measured the area of each photographed individual using the same batch processing in ImageJ with adjusted threshold and scale values. We then calculated the mean area for one *Gammarus* individual using the weighted mean of the area, which was obtained by calculating the median area of each photographed species and weighing it by species abundance. The obtained bio-area of *Gammarus* was then added to the macroinvertebrate bio-area of the respective samples. To compute the weighted mean area of

a *Niphargus* specimen we used the same approach and included 20 *Niphargus* specimens (2 *N. auerbachii*, 5 *N. luchoffmanni*, 2 *N. puteanus*, 3 *N. rhenorhodanensis* FG, 4 *N. thienemanni*, and 4 *N. tonywhitteni*). The obtained bio-area of *Niphargus*, *Gammarus*, and oligochaetes was assigned light or dark pigmentation based also on the specimen pictures.